BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

SQL: SELECT, FROM and WHERE

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Why close a file after use?

2. Which link best suits Bluetooth?
   A) Nearby wireless keyboard
   B) School buildings far apart
   C) Undersea Internet cable
   D) Long-distance fibre backbone

3. "Brute force works only by physically breaking the computer." Correct this. Explain why.

4. A network shows 2001:db8::7. "IPv6 uses four decimal numbers separated by dots." Correct this. How is this address written?

Today's objectives

Write simple SQL queries to retrieve records. Use the technique in a different problem.

Key vocabulary - Read, cover, recall

SELECT
Specifies columns required from a database table.

FROM
Identifies the database table from which records are selected.

WHERE
Filters records using a condition.

SQL query
SELECT Name FROM Pupils WHERE Year = 10 retrieves relevant names from matching rows.

BLACK PEN: Write one definition in your own words.

Do Now — Recall previous learning | **I Do** — Read new knowledge | **We Do** — Practise with guidance | **You Do** — Work independently | **Review & Improve** — Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

SELECT
Specifies columns required from a database table. In the school club database, identify where this technique applies.

FROM
Identifies the database table from which records are selected. In the library books database, identify where this technique applies.

WHERE
Filters records using a condition. In the sports team records, identify where this technique applies.

SQL query
SELECT Name FROM Pupils WHERE Year = 10 retrieves relevant names from matching rows. In the museum member list, identify where this technique applies.

I Do - Worked model

A school club database uses select and from. Explain what each contributes. [4 marks]

- Identify the program's purpose and the pieces of data needed.
- SELECT [1]; Specifies columns required from a database table. [1].
- FROM [1]; Identifies the database table from which records are selected. [1].
- Each feature needs a correct technical fact plus an explanation.

We Do - Guided practice

1. A library books database needs select. Explain the purpose. [2 marks]

2. Explain from for a sports team records. [2 marks]

3. When programming a museum member list, explain why where and sql query matter. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete this SQL query to return names of pupils with a mark greater than 60.

```
SELECT Name FROM Pupils WHERE Mark ... 60
```

You Do - Independent exam questions

Q2. State TWO relevant ideas or operations from this topic. [2 marks]

Q3. Explain from and where in a café inventory. [4 marks]

Q4. A programmer makes a school house records. Explain the roles of select, where and sql query. [6 marks]

Exit Ticket - True or False?

The SQL SELECT clause specifies which columns are retrieved from a database table.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It releases the handle or resources and helps ensure buffered writes are completed.

2. A - Nearby wireless keyboard

3. It tries many possible passwords or keys until one works.

4. IPv6 uses hexadecimal groups separated by colons. Groups of zeros can be shortened. Routers use IP addresses between networks; switches use MAC addresses within a LAN.

I Do - Explain the model

Identify the program's purpose and the pieces of data needed.

SELECT [1]; Specifies columns required from a database table. [1].

FROM [1]; Identifies the database table from which records are selected. [1].

Each feature needs a correct technical fact plus an explanation.

We Do - Model answers

1. SELECT [1]; Specifies columns required from a database table. [1].

2. FROM [1]; Identifies the database table from which records are selected. [1].

3. WHERE [1]; Filters records using a condition. [1]. SQL query [1]; SELECT Name FROM Pupils WHERE Year = 10 retrieves relevant names from matching rows. [1].

Code First - Answer

SELECT Name FROM Pupils WHERE Mark > 60
Expected output: Ada, Kai (example Pupils rows: Ada 62; Ben 45; Kai 80)

You Do - Self-mark

Q2: SELECT [1]; FROM [1].

Q3: FROM [1]; Identifies the database table from which records are selected. [1]. WHERE [1]; Filters records using a condition. [1].

Q4: SELECT [1]; Specifies columns required from a database table. [1]. WHERE [1]; Filters records using a condition. [1]. SQL query [1]; SELECT Name FROM Pupils WHERE Year = 10 retrieves relevant names from matching rows. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.